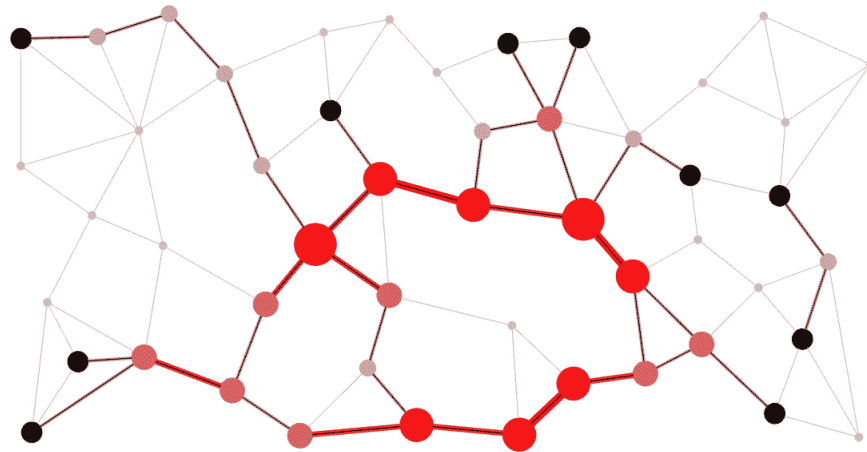


Can AI Learn Dynamics?

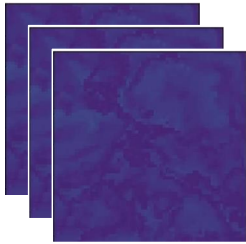
Surrogate Modeling on TACC Systems



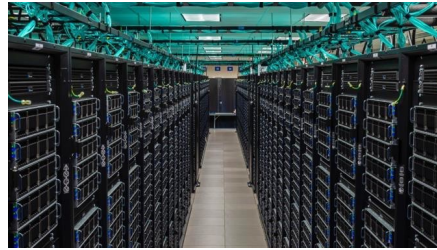
Luke Smith
Research Associate
Texas Advanced Computing Center
September 21, 2026

What does surrogate modeling look like on TACC systems?

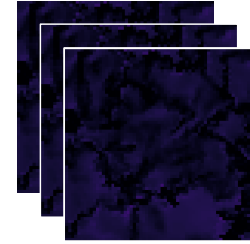
Initial Conditions



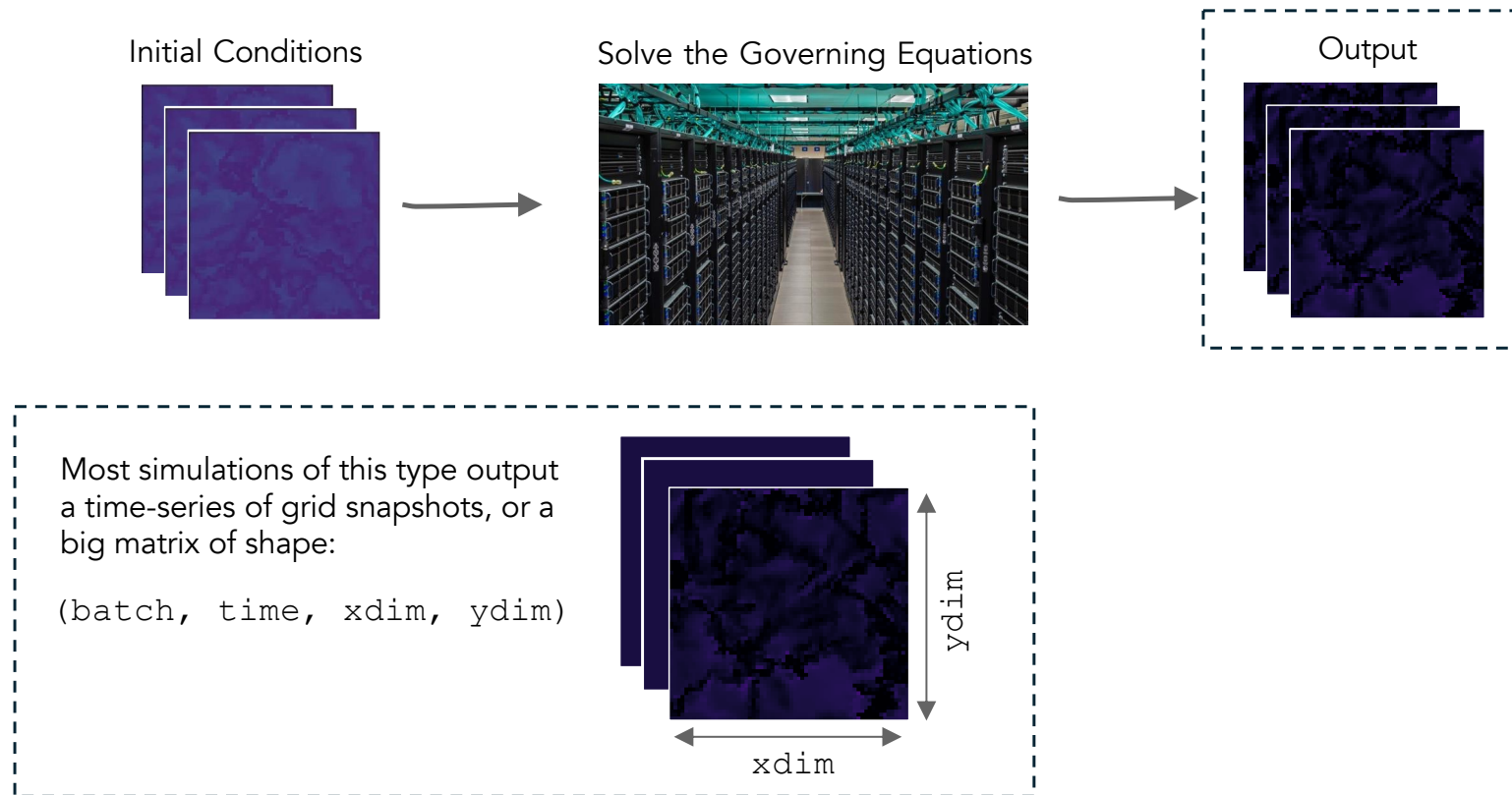
Solve the Governing Equations



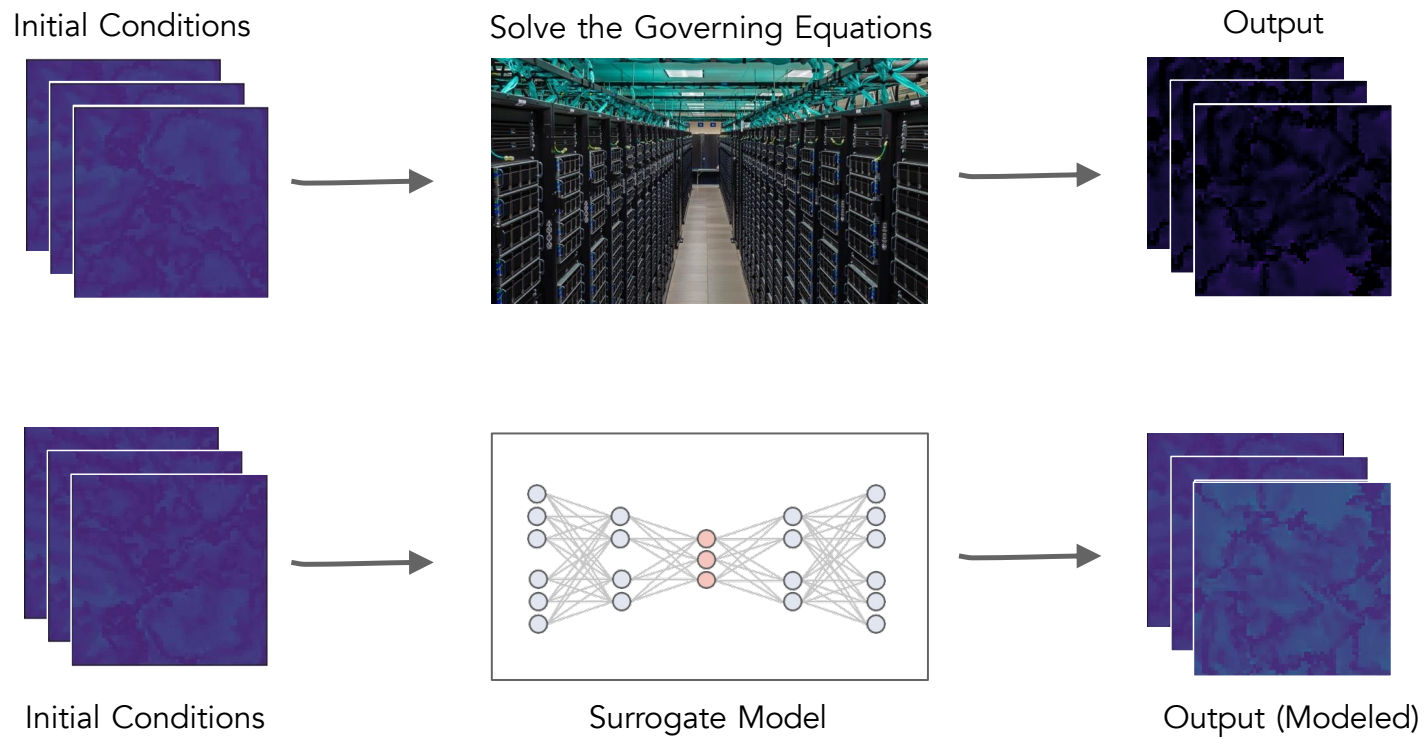
Output



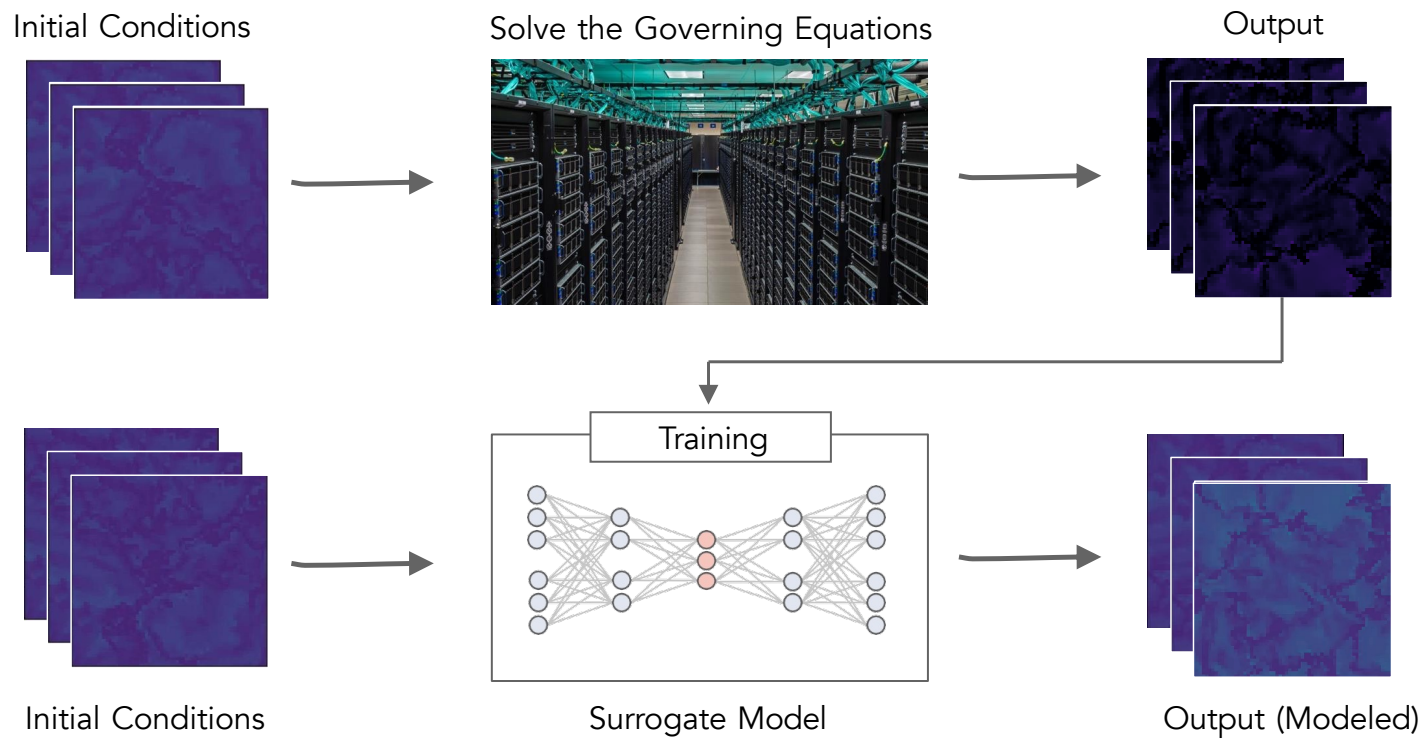
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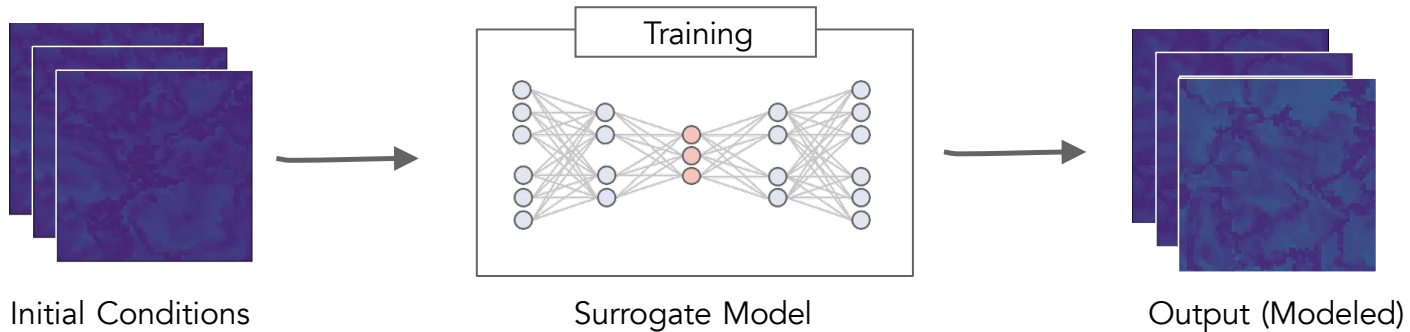
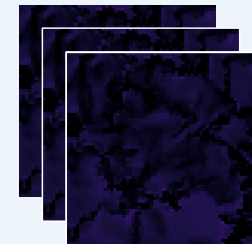


What does surrogate modeling look like on TACC systems?

How does this pipeline map to TACC systems?

- Generate training data: CPU Clusters on Stampede3, LS6, Vista, and Horizon
- Store Training data: Corral, Ranch, user directories on compute systems
- Train neural network: GPU resources on LS6, Vista, and Horizon
- Deployment: GPU resources on LS6, Vista, and Horizon

Training Data



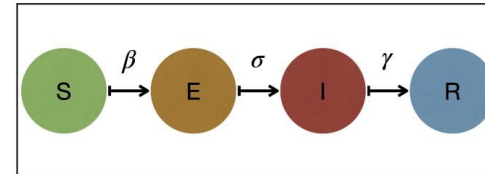
Tutorial Outline

What we'll cover

This tutorial provides a step-by-step walkthrough of setting up a surrogate modeling **pipeline** on TACC systems. We cover the following steps:

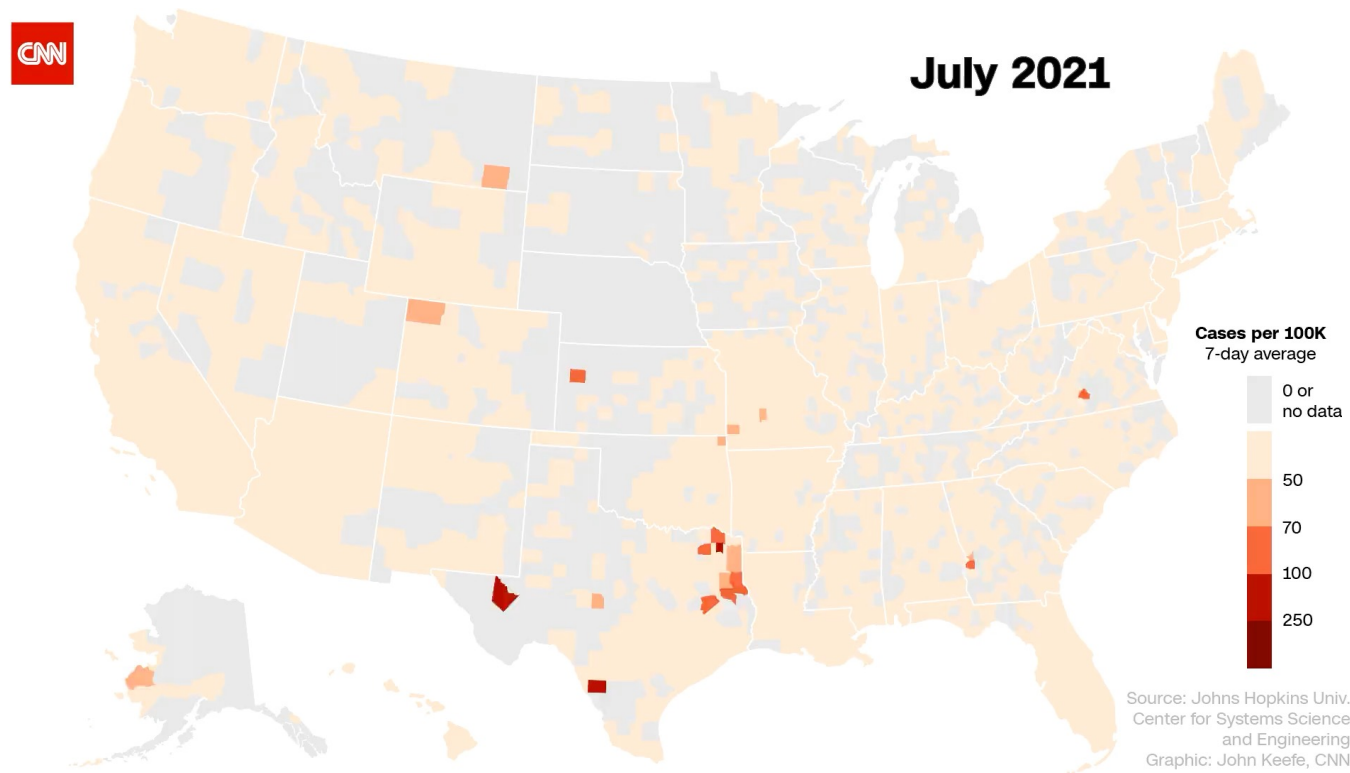
1. Data Loading
2. Data Normalization
3. Selecting an Architecture
4. Selecting an Objective
5. Training the Network
6. Testing/Evaluating the Model

What we'll take away



- Basic intuition for how TACC resources can be leveraged for surrogate modeling.
- A **template code** that can serve as a starting point for their own surrogate modeling pipeline.

Example: Disease Spread in the Epidemic Regime



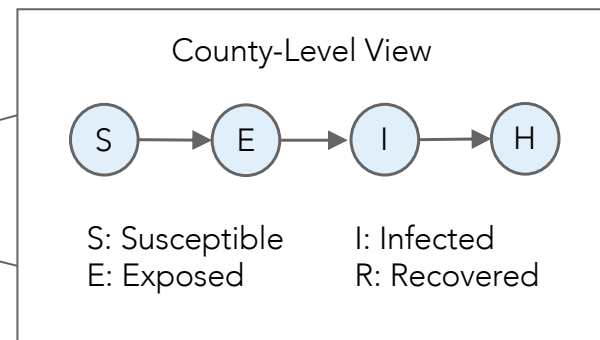
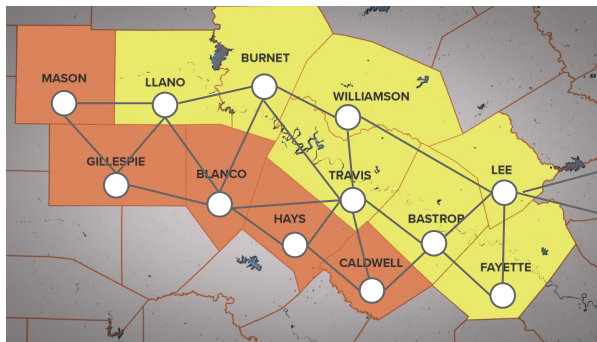
Disease Spread as a Dynamical System

In a modeling context, disease spread is formulated as a dynamical system in time:

$$\vec{x}_{n+1} = f(\vec{x}_n)$$

In this equation, $\vec{x}$ is the state vector, and f is a function that maps $\vec{x}$ from time n to time $n + 1$.

In the case of disease spread, the state vector $\vec{x}$ is stratified based on county, tracking a certain number of **compartments** per county.



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$$\vec{x}_n = \begin{bmatrix} S_n^{(1)} & E_n^{(1)} & I_n^{(1)} & R_n^{(1)} & S_n^{(2)} & E_n^{(2)} & \dots \end{bmatrix}$$

For an **SEIR compartment model**, the state vector $\vec{x}$ consists of the number of susceptible, exposed, infected, and recovered in each county (the superscript corresponds to county ID).

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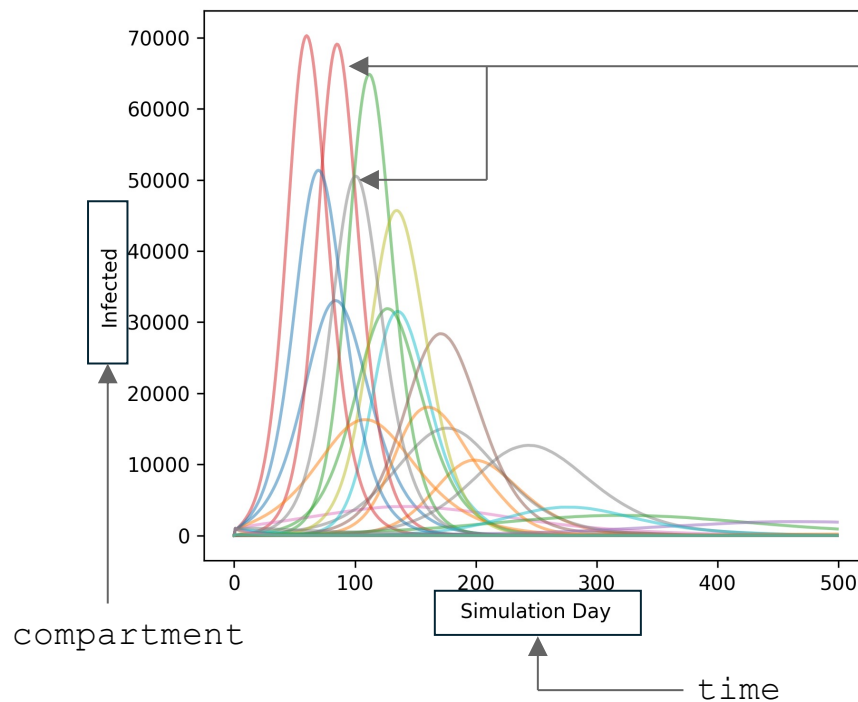
The temporal evolution of the state vector is simulated by a set of ordinary differential equations. These equations are solved per county, with coupling introduced by hyper-parameters β , σ , and γ .

$\frac{dS}{dt} = -\frac{\beta I}{N} S$	$\frac{dE}{dt} = \frac{\beta I}{N} S - \sigma E$	$\frac{dI}{dt} = \sigma E - \gamma I$	$\frac{dR}{dt} = \gamma I$
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In advanced simulations of disease spread, these equations are augmented with a parallel stochastic process that injects uncertainty into the state update. For purposes of this tutorial, we will focus on the **deterministic** version of the SEIR equations shown above.

Disease Spread as a Dynamical System

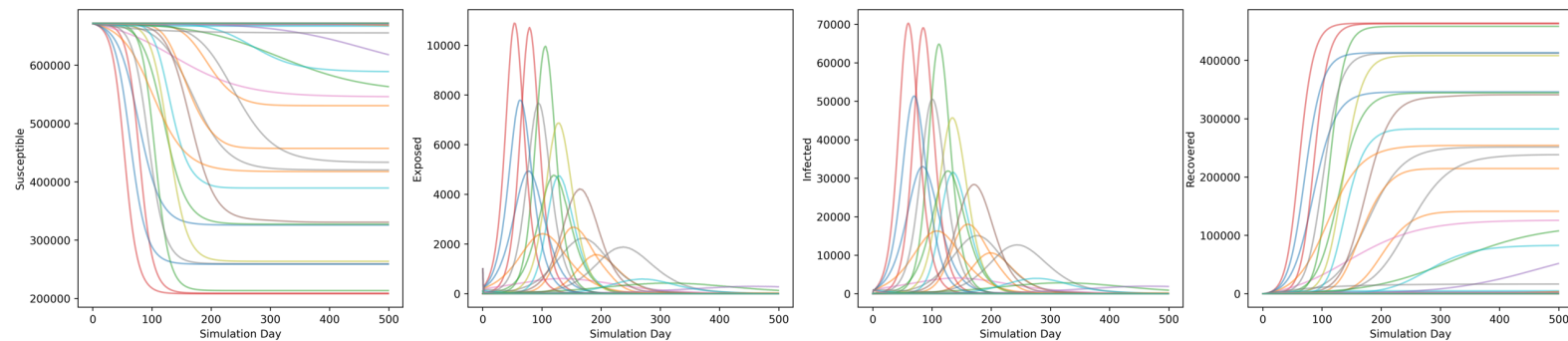
SEIR compartment simulations produce data that looks like this:



- Each line corresponds to a different initial condition (i.e., initial state vector and simulation constants β , σ , and γ).
- Following each line, one can trace the evolution of the epidemic: the infection begins, rises to a peak, and smoothly decays.
- The complete output dataset includes one of these plots for each **compartment** and **county** included in the simulation.

Tutorial: Surrogate Modeling in the Epidemic Regime

As a small, representative dataset, this tutorial considers 40 simulations of disease spread in Washington DC, which consists of only a single county/node. Each simulation corresponds to a different initial conditions, sweeping through the number of initial infected and the rate of infection β .



When viewed as a tensor, the complete dataset takes the following shape:

data_out: (40, 500, 4)
 ↑ ↑ ↑
 samples time SEIR

Tutorial: Surrogate Modeling in the Epidemic Regime

Our goal is to use this dataset to train a surrogate model, effectively distilling the dataset into a simple dynamics model. The surrogate allows for faster retrieval/interpolation for decision makers, while also providing an optimization framework for incorporating **measured** compartment data.

We design a surrogate model that maps each initial condition to the sample's full trajectory in time:

